

Eulerian Subgraphs

Time limit: 1.00 s

Memory limit: 512 MB

You are given an undirected graph that has n nodes and m edges.

We consider subgraphs that have all nodes of the original graph and some of its edges. A subgraph is called *Eulerian* if each node has even degree.

Your task is to count the number of Eulerian subgraphs modulo $10^9 + 7$.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines that describe the edges. Each line has two integers a and b : there is an edge between nodes a and b . There is at most one edge between two nodes, and each edge connects two distinct nodes.

Output

Print the number of Eulerian subgraphs modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
4 3 1 2 1 3 2 3	2

Explanation: You can either keep or remove all edges, so there are two possible Eulerian subgraphs.